

Diffusion Prior-Based Amortized VI for Noisy Inverse Problems

Sojin Lee, Dogyun Park, Inho Kong, Hyunwoo J.
Kim — Korea University

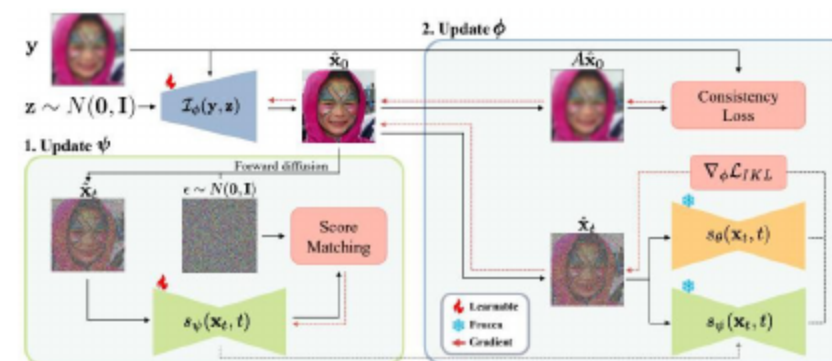
Intro

Noisy Inverse Problems & DAVI

3 Background and Motivation

- Noisy inverse problems pervade imaging and restoration.
- Model: $y = Ax + n$ with degradation and Gaussian noise.
- Ill-posedness makes recovery unstable and ambiguous.
 - Pretrained diffusion priors aid restoration, yet compute-heavy.
 - Per-instance and plug-and-play methods are slow, unscalable.
- Motive: amortized VI with diffusion priors for fast, robust inference.

4 Proposed Approach: DAVI Overview



- Freeze a powerful diffusion prior as the generator
 - Learn an amortized posterior conditioned on y
 - Train by minimizing KL to the true posterior
- Use score-matching and consistency training signals
 - Enforce measurement-model consistency at train/test
 - Enable single-step sampling for seen and unseen data

Sec 2

Related Works: Diffusion & AVI

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Related Works and Preliminary Foundations

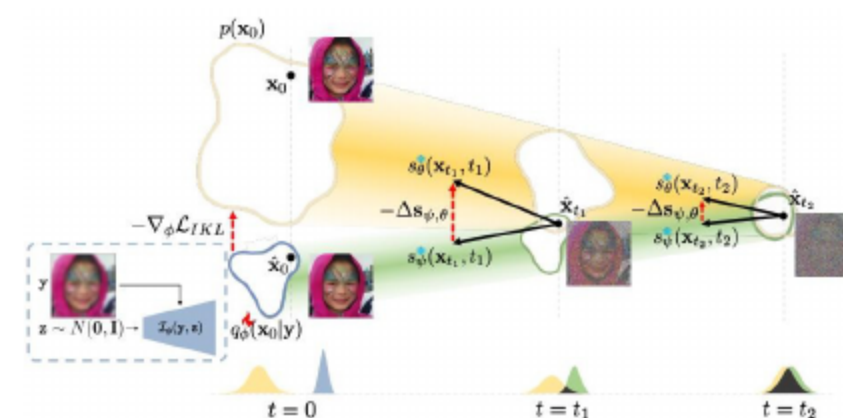
- Score-based, plug-and-play, posterior sampling priors.
- Methods like DDRM, DDNM show promise but are costly.
- Noise robustness and runtime remain key limitations.
 - Amortized VI learns variational posteriors via a network.
 - One-shot inference reduces per-instance optimization cost.
- Forward noise and reverse score nets underpin DAVI's prior.

Sec 3

Proposed Method: DAVI

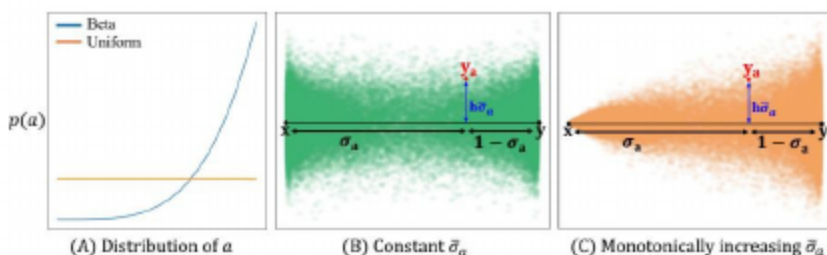
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DAVI: Components & Diffusion Posterior



- Amortized encoder maps measurements to posteriors
 - Frozen diffusion prior coupled with measurement model
 - Single-step sampling via implicit posterior mapping
- Training enforces prior- and data-consistency over time
 - Integral KL avoids support-mismatch divergences
 - Perturbed Posterior Bridge improves generalization

9 SDG and Perturbed Posterior Bridge

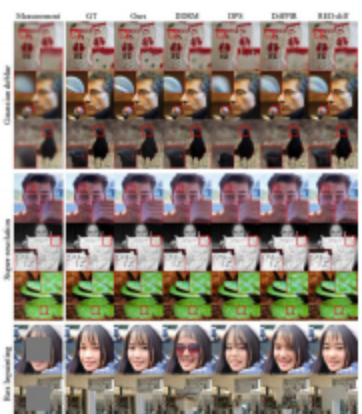


- SDG approximates IKL gradients via score functions
 - Aligns amortized posterior with diffusion prior
 - Alternating updates improve convergence stability
- PPB inserts perturbed intermediate trajectories
 - Regularizes posterior paths under noise mismatch
 - Boosts fidelity and generalization to new data

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Experiments

2 Results: Deblur, SR, Inpainting



- Higher PSNR/SSIM; sharper textures, fewer artifacts
 - Lower FID and LPIPS across deblur, SR, inpainting
 - Robust to varied noise levels; maintains realism
- Amortization cuts function evals; 2-10x faster

12 Conclusion and Takeaways

- DAVI unifies diffusion priors and amortized VI with neural implicit.
- Delivers fast, accurate, robust reconstructions, surpassing baselines.
- SDG with a frozen diffusion prior stabilizes training.
 - Perturbed posterior bridge generalizes across noisy inverse problems.
 - Future: broader measurements, UQ, higher resolutions.
- Ethical use: datasets and resemblance risks noted.